Atiksh Bhardwaj

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Education

Carnegie Mellon University, Robotics Institute at School of Computer Science — Pittsburgh, PA
Master's of Science in Robotics **[Supported by NSF GRFP]** — August 2026-Present

Cornell University, College of Computer and Information Sciences — Ithaca, NY
Bachelor of Science in Computer Science and Mathematics with Honors — August 2022-December 2025
GPA: 3.97/4.00

Honors and Awards: Dean's Honors List and Honors Program [Requires external research and graduate courses]
Related Courses: Deep Learning for Robotics, Numerical Analysis of Differential Equations, Algorithms, Reinforcement Learning, Machine Learning, Computer Vision, Artificial Intelligence, Systems and Organization, Discrete Math

Technical Skills

- **Back-end Development Tools:** Python, Java, OCaml, MATLAB, Simulink, ROS C, C++, SQL
- **Front-end Development Tools:** Electron, HTML, CSS, JavaScript, Qt
- **Developer Tools:** Git, GitHub, VS Code, PyTorch, Numpy, TensorFlow, OpenCV, JAX, Claude Code, Codex
- **Cloud Development Tools:** AWS, Azure, Google Cloud
- **Computer:** Linux, Windows, MacOS Microsoft Suite, Blender, CUDA, LLMs, Docker

Publications | Google Scholar

Progress Probes P^2: Lightweight Runtime Monitoring for Vision-Language-Action Models (coming soon)
Atiksh Bhardwaj, Edward W. Duan, Junwon Seo, et. al, Andrea Bajscy†, Preston Culberston†
Currently **Under Review**

Decoding Task Progress from VLA Representations [arxiv]
Atiksh Bhardwaj*, Edward W. Duan*, Prithwish Dan, Wei-Chiu Ma, Preston Culberston | **Preprint**

X-Diffusion: Training Diffusion Policies on Cross-Embodiment Human Demonstration [arxiv] [website] [code]
Maximus A. Pace*, Prithwish Dan*, Chuanruo Ning, **Atiksh Bhardwaj**, et. al, Wei-Chiu Ma†, Kushal Kedia†
Accepted at **ICRA 2026**

A Smooth Sea Never Made a Skilled SAILOR: Robust Imitation via Learning to Search [arxiv] [website] [code]
Arnav Kumar Jain*, Vibhakar Mohta*, Subin Kim, **Atiksh Bhardwaj**, et. al, Sanjiban Choudhury, Gokul Swamy
Published as **Spotlight Paper at NeurIPS 2025** (Top 3% of papers; 688 selected out of 21544 submissions)

MOSAIC: A Modular System for Assistive and Interactive Cooking [arxiv] [website] [thread]
Yuki Wang*, Kushal Kedia*, Juntao Ren*, Rahma Abdullah, **Atiksh Bhardwaj**, et. al, Sanjiban Choudhury
Published at **CoRL 2024**
Best Paper at ICRA 2024 VLNMN Workshop and Best Poster at ICRA 2024 MoMa Workshop

InteRACT: Transformer Models for Human Intent Prediction Condition on Robot Action [arxiv] [website] [code] [data]
Kushal Kedia, **Atiksh Bhardwaj**, Prithwish Dan, Sanjiban Choudhury | Published at **ICRA 2024**

ManiCast: Collaborative Manipulation with Cost-Aware Human Forecasting [arxiv] [website] [code] [data]
Kushal Kedia, Prithwish Dan, **Atiksh Bhardwaj**, Sanjiban Choudhury | Published at **CoRL 2023**

Experience

Graduate Researcher — May 2026-Present
- Advised by Prof. Andrea Bajcsy at the Intent Robotics Lab at CMU doing interpretability research
- Funded via NSF GRFP and completed a research assistantship at Intent Lab during Summer 2026

Research Assistant — January 2026-May 2026
- Advised by Prof. Preston Culbertson at the PRAXIS Lab at Cornell University doing interpretability research

Undergraduate Robotics and Machine Learning Researcher — May 2023-December 2025
- Advised by Prof. Sanjiban Choudhury at PoRTaL Lab at Cornell University
- Bowers Undergraduate Research Experience during summers 2023/2024 for a **$5000/$7000 fellowship**

Texas Instruments Software Engineering Internship — May-August 2025
- Optimized CNN, RNN, and LSTM based Audio AI on TI Sitara processors via open-source TVM code generation
- Improved performance in code generation by 13x and nearly 4x versus competitor and open-source hardware

Brains4Drones Internship — May-August 2022
- Created automatic panorama stitcher to make LIDAR maps through OpenCV and C++ based on drone movement
- Coded GUI application in JavaScript for line workers to view footage and automated Lidar Map in real-time